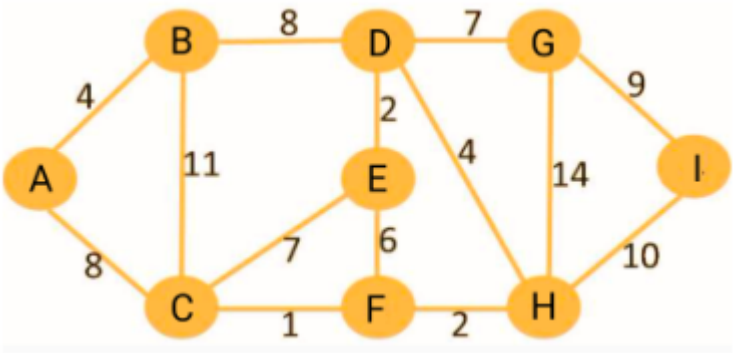
Problem 1 [30 pts]

Find the shortest path from source node A to all the other nodes in the following graph. Use Dijkstra's algorithm by following the tabular approach as we have discussed in class (example in the lecture slides). For each shortest path, list the nodes in order, along with its corresponding total distance.



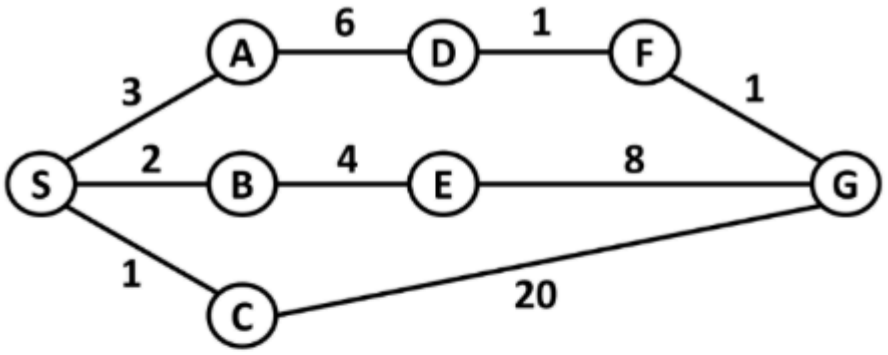
node	start	A(0)	B(4)	C(8)	F(9)	H(11)	D(12)	E(14)	G(19)	I(21)
A	0									
B	∞	4								
C	∞	8								
D	∞		12							
E	∞			15			14			
F	∞			9						
G	∞					25	19			
H	∞				11					
I	∞					21				

A(0) → B(4) → C(8) → F(9) → H(11) → D(12) → E(14) → G(19) → I(21)

Destination	Shortest Path	Total Distance
B	A → B	4
C	A → C	8
D	A → B → D	12
E	A → B → D → E	14
F	A → C → F	9
G	A → B → D → G	19
H	A → C → F → H	11
I	A → C → F → H → I	21

Problem 2 [70 pts]

Given the following graph where the vertex S is the start state (or initial state) and the vertex G is the goal state, use A* algorithm to find the shortest path.



Use this heuristic function:

n	$h(n)$
A	7
B	10
C	15
D	1
E	8
F	1
G	0
S	10

A* algorithm:

Path	g(n)	h(n)	f(n) = g(n) +h(n)	Shortest
S → A	3	7	10	10
S → B	2	10	12	
S → C	1	15	16	
S → A → D	3 + 6 = 9	1	10	10
S → A → D → F	3 + 6 + 1 = 10	1	11	11
S → A → D → F → G	3 + 6 + 1 + 1 = 11	0	11	11

Vertices explored: S → A → D → F → G

Shortest path: S → A → D → F → G

Total cost = 11

Check Admissibility:

A heuristic is admissible if: h(n) ≤ true cost t(n)

n	h(n)	True cost t(n) to G	Admissibility
S	10	S → A → D → F → G = 11	yes
A	7	A → D → F → G = 8	yes
B	10	B → E → G = 12	yes
C	15	C → G = 20	yes
D	1	D → F → G = 2	yes
E	8	E → G = 8	yes
F	1	F → G = 1	yes
G	0	0	yes

Conclusion: The heuristic is admissible because every h(n) value is less than or equal to the real shortest cost to G.

Check Consistency:

A heuristic is consistent if: $h(n) \leq c(n,n') + h(n')$

$n \rightarrow n'$	$h(n)$	$c(n,n') + h(n')$	Consistent
$S \rightarrow A$	10	3 + 7 = 10	yes
$S \rightarrow B$	10	2 + 10 = 12	yes
$S \rightarrow C$	10	1 + 15 = 16	yes
$A \rightarrow D$	7	6 + 1 = 7	yes
$B \rightarrow E$	10	4 + 8 = 12	yes
$C \rightarrow G$	15	20 + 0 = 20	yes
$D \rightarrow F$	1	1 + 1 = 2	yes
$E \rightarrow G$	8	8 + 0 = 8	yes
$F \rightarrow G$	1	1 + 0 = 1	yes

Conclusion: The heuristic is consistent because every connected pair follows $h(n) \leq c(n,n') + h(n')$.

Is A* Optimal?

Yes. Since the heuristic is admissible and consistent, A* gives the shortest path.

The shortest path is $S \rightarrow A \rightarrow D \rightarrow F \rightarrow G$, and the total cost is 11.